Part A: Filtering & Edge Detection

Image Filtering (图像滤波)

1. **Linear Filtering (线性滤波)**
   - **Correlation (互相关)** $G = H \otimes F$: $G[i,j] = \sum_{u,v} H[u,v]F[i+u, j+v]$.
   - **Convolution (卷积)** $G = H * F$: Same as correlation but kernel H is **flipped** (horizontally & vertically).
   - **Properties**: Commutative ($f * g = g * f$), Associative.
   - **Gaussian Filter (高斯滤波)**:
     - Formula: $G_\sigma(x,y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$.
     - Role: **Low-pass filter**, removes high-freq noise, blurs image.
     - Weights: Sum of kernel elements ≈ 1 (Normalization).

Coach's Past Paper 考法解析

考点/陷阱 (Teacher's Notes):
- **Kernel Size vs** σ: $\uparrow \sigma \Rightarrow$ More blur $\Rightarrow$ Detect large scale features. $\downarrow \sigma \Rightarrow$ Less blur $\Rightarrow$ Detect fine details.
- **Trap**: Convolution requires **flipping** the kernel. If kernel is symmetric (like Gaussian), Convolution = Correlation.

2. **Non-linear Filtering (非线性滤波)**
   - **Median Filter (中值滤波)**:
     - Method: Sort neighbors, pick the **median** (50% rank).
     - **CRITICAL**: Best for removing **Salt & Pepper Noise (椒盐噪声)** while **preserving edges (保留边缘)**.
     - Not linear: $Median(A + B) \neq Median(A) + Median(B)$.

Edge Detection (边缘检测)

1. **Gradient & Derivatives**
   - **Goal**: Identify visual changes (discontinuities).
   - **Gradient Vector**: $\nabla f = [\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}]^T$. Points to direction of max intensity increase.
   - **Magnitude (幅值)**: $||\nabla f|| = \sqrt{f_x^2 + f_y^2}$ (Edge Strength).
   - **Direction (方向)**: $\theta = \tan^{-1}(f_y/f_x)$ (Edge Normal).
   - **Finite Difference**: $f_x \approx F[x+1] - F[x]$. Kernel: $[-1, 0, 1]$.
   - **Sobel Operator**: Combines smoothing + differentiation.
2. **Laplacian of Gaussian (LoG)**
   - Uses 2nd derivative: $\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$.
   - **Zero-crossing**: Edges are found where LoG crosses zero (inflection point).
   - Shape: "Mexican Hat" (Sombrero).
   - Formula: $\nabla^2(G * f) = (\nabla^2 G) * f$. (Associative property saves computation).
3. **Canny Edge Detector (Canny 算子)**

   The optimal edge detector. Memorize the 4 steps!

   a. **Gaussian Smoothing**: Apply Gaussian filter to reduce noise.

   b. **Gradient Calculation**: Compute Magnitude (M) and Orientation (θ) at each pixel.

   c. **Non-Maximum Suppression (NMS)**:
   - Goal: **Thinning (细化)** edges to 1-pixel width.
   - How: Check pixel P against neighbors **along the gradient direction**. If $P <$ neighbors, set $P = 0$ (suppress).

   d. **Hysteresis Thresholding (双阈值)**:
   - Define T_{high} and T_{low}.
   - **Strong Edge**: $M > T_{high}$ (Keep).
   - **Weak Edge**: $T_{low} < M < T_{high}$. Keep **ONLY IF** connected to a strong edge.
   - **Noise**: $M < T_{low}$ (Discard).

必考 (Exam Focus)
- **Why Smooth?** Derivative is high-pass, amplifies noise. Must smooth first!
- **NMS Details**: Past paper Q1(a) asked about NMS application. It ensures good localization (single response).
- **Comparison**: 1^{st} Derivative $\rightarrow$ Look for **Peaks/Maxima**. 2^{nd} Derivative (LoG) $\rightarrow$ Look for **Zero-crossings**.

Part A Supplement: 核心概念大白话讲解 (考场救急用)

Image Filtering (图像滤波 - 怎么让图变干净?)

- **目标**: 去除噪点 (Denoising)，就像给图片“磨皮”。
- **Linear Filtering (线性滤波)**:
  - **动作**: 拿个小方框 (Kernel) 在图上滑，算加权平均。
  - **Convolution vs Correlation**:
    - ⋆ **唯一区别**: 卷积 (Convolution) 要先把方框 **上下左右翻转 (Flip)** 再算；互相关不翻转。
    - ⋆ **考点**: 看到"Convolution"，找关键词 **"flipped"**。
  - **Gaussian Filter**: 最常用的“高斯模糊”，用来平滑图像，是 **Low-pass filter** (低通)。
- **Non-linear Filtering (非线性滤波) - 必考**:
  - **Median Filter (中值滤波)**:
    - ⋆ **原理**: 像素排队，选**中间**那个。
    - ⋆ **必杀技**: 专治 **"Salt & Pepper Noise" (椒盐噪声)**，而且**不模糊边缘** (Preserving edges)。
    - ⋆ **考点**: 题目有"Salt & Pepper"，必选 Median。

Edge Detection (边缘检测 - 怎么描边?)

- **目标**: 找到颜色剧烈变化的地方 (梯度大)。
- **Gradient (梯度)**: 变化率。
  - **一阶导**: 找波峰 (Peak)。
  - **二阶导 (LoG)**: 找 **"Zero-crossing" (过零点)**。
- **Canny Edge Detector (Canny 算子) - 必背 4 步**:
  (a) **Gaussian Smoothing**: 先磨皮去噪 (因为导数怕噪声)。
  (b) **Gradient Calculation**: 算哪里变化大 (幅值) 和方向。
  (c) **NMS (非极大值抑制)**: 把粗边变细。关键词: **Thinning (细化)**。
  (d) **Hysteresis (双阈值)**: 用高低两个门槛连线，不断不乱。

考场生存指南 (How to use Cheat Sheet)

- **选择题 (Multiple Choice)**:
  - "Salt & Pepper" $\rightarrow$ 找 **Median Filter**。
  - "Thin edges?" $\rightarrow$ 找 **NMS**。
  - "Zero-crossing" $\rightarrow$ 找 **LoG**。
  - "Convolution vs Correlation" $\rightarrow$ 找 **"flipped"**。
- **简答题**:
  - 问 **Canny 步骤**? $\rightarrow$ 抄上面的 4 步。
  - 问 **为啥先 Gaussian**? $\rightarrow$ 答"Derivative amplifies noise" (导数放大噪声，必须先降噪)。

Part A 计算题攻略 (Calculation Guide):
- **1. Linear Filtering (Convolution)**:
  - **Case 1 (Symmetric Kernel)**: Multiply corresponding pixels and sum up. (e.g., Gaussian, Mean, Laplacian).
  - **Case 2 (Asymmetric Kernel)**: **Flip** the kernel (rotate 180°) first! Then multiply and sum.
- **2. Median Filter**:
  - List all pixels in the window (e.g., 3x3 $\rightarrow$ 9 pixels).
  - **Sort** them: $v_1 \leq v_2 \leq ... \leq v_9$.
  - Pick the **middle** one (v_5). NO math calculation involved!
- **3. Gradient**:
  - $G_x \approx I(x+1, y) - I(x-1, y)$ (Right - Left)
  - $G_y \approx I(x, y+1) - I(x, y-1)$ (Bottom - Top)
  - Mag = $\sqrt{G_x^2 + G_y^2}$, Dir = $\tan^{-1}(G_y/G_x)$.

Part B: Sampling & Aliasing

Sampling Theory (采样理论)

1. **Nyquist Theorem (奈奎斯特采样定理)**
   - To reconstruct perfectly, sampling rate f_s must satisfy:
     $$f_s > 2 \cdot f_{max}$$
     where f_{max} is the highest frequency in the signal.
2. **Aliasing (混叠现象)**
   - Occurs when $f_s \leq 2 f_{max}$ (Under-sampling).
   - **Frequency Domain**: Sampling causes the spectrum to **replicate** at multiples of f_s. Aliasing is when these replicas **overlap**.
   - Result: High-freq artifacts (Moire patterns, jagged edges).
3. **Anti-aliasing (抗混叠)**
   - **Solution**: Apply a **Low-pass Filter (Prefilter)** (e.g., Gaussian) **BEFORE** sampling to remove frequencies $> f_s/2$.

- **Ideal Filter**: Sinc function (Box in frequency domain), but it's infinite/unrealizable.

Image Resizing (图像缩放)

1. **Down-sampling (降采样)**
   - **Wrong**: Decimation (keeping every k^{th} pixel) $\rightarrow$ Aliasing.
   - **Right**: **Blur (Gaussian)** $\rightarrow$ **Subsample**.
   - Matlab: `imresize` automatically does anti-aliasing (unless using 'nearest').
2. **Up-sampling (升采样)**
   - Requires **Interpolation** to estimate unknown pixels.
   - **Inverse Warping**: Map dest (x', y') to source (x, y).

Interpolation Methods (插值)

1. **Nearest Neighbor (最近邻)**
   - Method: $f(x) = f(\text{round}(x))$.
   - Kernel: Box / Rectangle function.
   - Freq Domain: Sinc function (bad low-pass filter).
   - **Cons**: **Blocky artifacts (马赛克)**, jagged edges.
2. **Bilinear Interpolation (双线性)**
   - Method: Weighted avg of **4 nearest neighbors**.
   - Kernel: Triangle / Tent function.
   - Freq Domain: Sinc^2 function.
   - **Cons**: Blurs fine details.
3. **Bicubic (双三次)**
   - Method: Uses **16 neighbors**.
   - Better approximation of Sinc. Sharper than Bilinear.

> **Coach's Past Paper 考法解析**
>
> **重点 (Revision Focus):**
> - **Pre-filtering**: Teacher emphasized "MUST pre-filter before downsampling".
> - **Spectrum**: Sampling = Spectrum Replication. Aliasing = Spectrum Overlap.
> - **Calculation**: Inverse Warping usually gives float coordinates (e.g. 1.5). Nearest takes pixel at 2; Bilinear averages pixels at 1 and 2.

Part B Supplement: 采样与混叠核心讲解

Sampling Theory (采样理论 - 怎么变小?)

- **Nyquist Theorem (奈奎斯特 - 必背)**:
  - 规则: 采样频率必须 $>2\times$ 信号最高频率。
  - 公式: $f_s > 2 \cdot f_{max}$。
  - 没做到会怎样? $\rightarrow$ **Aliasing (混叠)** (如摩尔纹、车轮倒转)。
- **Anti-aliasing (抗混叠 - 必考流程)**:
  - 为了防止混叠，必须在缩小前先模糊。
  - **Step 1**: **Pre-filter** (Low-pass/Gaussian Blur) 去掉高频。
  - **Step 2**: **Subsample** (降采样)。

Interpolation (插值 - 怎么变大?)

- **场景**: 放大图片时，填补未知的像素值。
- **Nearest Neighbor (最近邻)**:
  - **方法**: 找最近的 **1** 个邻居。
  - **特点**: 有 **锯齿 (Jagged)** / **马赛克 (Blocky)**。
- **Bilinear (双线性)**:
  - **方法**: 找周围 **4** 个邻居加权平均。
  - **特点**: 平滑但 **模糊 (Blur)**。
- **Bicubic (双三次)**:
  - **方法**: 找周围 **16** 个邻居。
  - **特点**: 效果最好，最慢。

考场生存指南 (How to use)

- **选择题**:
  - "Condition to avoid aliasing?" $\rightarrow$ 找 **Nyquist** ($>2f_{max}$)。
  - "Steps to resize?" $\rightarrow$ 找 **Pre-filter then Sample**。
  - "Blocky artifacts?" $\rightarrow$ 找 **Nearest Neighbor**。
- **计算题 (Inverse Warping)**:
  - 如果算出原图坐标是小数 (如 1.5)，必须用 **Interpolation**。
  - 不要直接四舍五入 (除非题目让你用 Nearest Neighbor)。

> **计算题必胜攻略 (Calculation Guide):**
> - **Scenario**: Upsample image by $S = 2$. Find value at dest $(3, 5)$.
> - **Step 1: Inverse Warping (找原坐标)**
>
> $$u = x'/S = 3/2 = 1.5, \quad v = y'/S = 5/2 = 2.5$$
>
> - **Step 2: Interpolation (算数值)**
>   - **Nearest Neighbor**: Round $(1.5, 2.5) \rightarrow (2, 3)$. Take $I(2, 3)$.
>   - **Bilinear**: Let $a = 0.5, b = 0.5$.
>
> $$Val = (1-a)(1-b)I(1,2) + a(1-b)I(2,2) + ...$$
>
> For midpoint (0.5), it is just **Average of 4 neighbors**:
>
> $$\frac{I(1,2) + I(2,2) + I(1,3) + I(2,3)}{4}$$

Part C: Color Spaces(颜色空间)

Color Spaces (颜色空间)

1. **RGB Color Space**
   - **Components**: Red, Green, Blue.
   - **Application**: Monitors, Cameras, Displays.
   - **Normalized**: $r = R/(R + G + B)$, removing intensity to some extent.
2. **HSI / HSV Color Space**
   - **Components**:
     - **Hue (色调)**: Dominant color (Angle $0 - 2\pi$).
     - **Saturation (饱和度)**: Purity of color (Radius).
     - **Intensity/Value (亮度)**: Brightness (Height).
   - **Pros**: Decouples intensity from color info. More intuitive to human perception.
   - **Application**: **Image Segmentation (图像分割)** based on color.
3. **YIQ Color Space**
   - **Components**:
     - **Y (Luminance)**: Brightness/Intensity (Gray-scale).
     - **I & Q (Chrominance)**: Color information.
   - **Application**: NTSC TV standard (backward compatibility with B&W TV).
   - **Relation**: Y is a weighted sum of R, G, B.

> **Coach's Past Paper 考法解析**
>
> **考点/陷阱 (Teacher's Notes):**
> - **Why HSI?**: RGB is not good for segmentation because color and intensity are mixed. HSI separates them, making it robust to lighting changes (using only H & S).
> - **Color Histogram**: Invariant to translation and rotation (2D), but sensitive to scale (if not normalized) and lighting.

Texture Analysis (纹理分析)

1. **Statistical Approaches (统计方法)**
   - **Edge Density**: Number of edge pixels per unit area.
   - **Co-occurrence Matrix (共生矩阵)**:
     - Counts how often pairs of pixels with specific value and spatial relationship $d = (dr, dc)$ occur.
     - **Trap**: Expensive to compute for many d.
2. **Local Binary Pattern (LBP)**
   - **Method**: Compare center pixel with neighbors.
   - If neighbor $\geq$ center, output 1; else 0. Concatenate bits to form a binary number.
   - **Pros**: Rotation invariant (if using uniform patterns), robust to monotonic gray-scale changes.
   - **Application**: Texture classification, Face recognition.
3. **Filter Banks (滤波器组)**
   - Use a bank of filters (e.g., Gabor, Gaussian derivatives) at different **scales** and **orientations**.
   - **Response Vector**: Collect responses from all filters at each pixel.

Part C: 补充详解

Color Spaces (颜色空间)

1. **RGB (Red, Green, Blue)**
   - **原理**: 类似显示器像素点，由红绿蓝三盏灯混合而成。

- **缺点**: **颜色 (Chrominance) 和亮度 (Intensity) 混在一起**。
- **举例**: 红苹果在阴影下 R, G, B 值都会大幅变小，计算机可能不再认为它是“红色”（不适合图像分割）。
- **考点**: 常用于 Display (显示) 和 Capture (拍摄)。

2. **HSI / HSV (Hue, Saturation, Intensity)**
   - **设计初衷**: 符合人眼直觉。
   - **组成**:
     - **Hue (色调)**: 颜色的种类（角度 $0-360^\circ$）。
     - **Saturation (饱和度)**: 颜色的纯度/鲜艳程度。
     - **Intensity (亮度)**: 光的强弱。
   - **优点 (必考)**: **解耦了亮度与颜色 (Decouple)**。不管亮度 (I) 怎么变，红色的 Hue 值基本不变。
   - **应用**: **Image Segmentation (图像分割)**，特别是在复杂光照条件下。
3. **YIQ**
   - **组成**:
     - **Y (Luminance)**: 亮度信息（黑白信号）。
     - **I & Q (Chrominance)**: 色度信息。
   - **优点**: 人眼对亮度 (Y) 敏感，对颜色 (I, Q) 不敏感，传输时可压缩 I/Q 带宽。且兼容黑白电视。
   - **考点**: 用于 **TV Broadcasting (电视广播)** (NTSC 标准)。

Texture (纹理)

1. **LBP (Local Binary Pattern) - 局部二值模式**
   - **原理**:
     A. 取 3×3 窗口，以中心像素为阈值。
     B. 邻居 $\geq$ 中心记 1，邻居 $<$ 中心记 0。
     C. 顺时针排列得到二进制数。
   - **考点**: **对光照变化鲁棒 (Robust to illumination)**。如果整体变亮（像素值 +10），相对大小不变，LBP 码不变。
2. **Co-occurrence Matrix (共生矩阵)**
   - **原理**: 统计“距离为 d、方向为 θ 的两个像素，灰度分别为 i 和 j”出现的次数。
   - **缺点**: 计算量大。

Calculations (计算题考法)

Coach's Past Paper 考法解析

考法 1: LBP 计算
- **题目**: 给定 3×3 矩阵，求中心像素 LBP 值。
- **矩阵**:

$$\begin{bmatrix} 5 & 9 & 1 \\ 4 & \mathbf{6} & 8 \\ 7 & 2 & 3 \end{bmatrix}$$

- **解法**: 中心是 6。

$$\begin{array}{ccc} 5<6(\mathbf{0}) & 9\geq6(\mathbf{1}) & 1<6(\mathbf{0}) \\ 4<6(\mathbf{0}) & \text{Center} & 8\geq6(\mathbf{1}) \\ 7\geq6(\mathbf{1}) & 2<6(\mathbf{0}) & 3<6(\mathbf{0}) \end{array}$$

- **结果**: 顺时针读数 (如从左上角起) $\rightarrow 01010010_2 \rightarrow$ 转十进制。

考法 2: 共生矩阵 (Co-occurrence Matrix) 填空
- **题目**: 给定小图像，填 $d=(1,0)$ (右边邻居) 的矩阵。
- **解法**: 数数。比如矩阵位置 $(0,0)$ 填几?
- **操作**: 去原图里数，有多少对相邻像素是”0 0” (左边是 0，右边也是 0)。

Part D: Deep Learning (深度学习)

Neural Networks (神经网络基础)

1. **Activation Functions (激活函数)**
   - **Sigmoid**: $\sigma(z)=\frac{1}{1+e^{-z}}$.
   - **ReLU (Rectified Linear Unit)**: $f(x)=\max(0,x)$.
   - **Comparison**: ReLU is preferred to avoid **Vanishing Gradient (梯度消失)** problem (because Sigmoid derivative ≤ 0.25).
2. **Training (训练)**
   - **Loss Function**: Cross-entropy (for classification).
   - **Backpropagation (反向传播)**: Uses **Chain Rule** to compute gradients $\frac{\partial L}{\partial w}$ for weight updates.

CNN (卷积神经网络)

1. **Convolutional Layer (卷积层)**
   - Uses learnable **Filters/Kernels** to extract local features.
   - **Feature Map**: The output of convolution. Preserves spatial arrangement.
2. **Pooling Layer (池化层)**
   - **Types**: Max Pooling (take max), Average Pooling (take mean).
   - **Advantages (Key Exam Point)**:
     (a) Reduces dimensionality & computation (降维/减参).
     (b) Provides **Invariance** (to translation/shift) (平移不变性).
3. **Regularization (正则化)**
   - **Dropout**: Randomly dropping units during training to prevent **Overfitting**.
   - **Batch Normalization**: Normalizes layer inputs to mean 0, var 1. Accelerates training.

Coach's Past Paper 考法解析

必考 (Exam Focus - Past Paper Q1/Q2):
- **Pooling Advantages**: Must mention 1. Computation reduction; 2. Invariance (Translation).
- **Receptive Field (感受野)**: The region in the input image that a pixel in the output feature map ”sees”.
- **Feature Map to Vector**: How to put a $13\times13\times256$ tensor into a classifier? **Ans**: Flattening or Global Average Pooling.
- **Trap**: CNN is NOT scale invariant by default (requires Multi-scale training or pyramids).

Part E: Segmentation & Object Detection (分割与检测)

Image Segmentation (图像分割)

1. **Semantic Segmentation (语义分割)**
   - **Goal**: Label each pixel with a class (e.g., sky, road, car).
   - **Key**: Does NOT distinguish instances (all ”cars” are the same label).
   - **Model**: **FCN (Fully Convolutional Network)**.
     - Replaces Fully Connected layers with Convolutional layers.
     - Uses **Upsampling / Deconvolution** to recover resolution.
2. **Instance Segmentation (实例分割)**
   - **Goal**: Detect and segment each **individual object** instance.
   - **Key**: Distinguishes ”Cat A” from ”Cat B”.
   - **Model**: **Mask R-CNN** (Faster R-CNN + Mask branch).

Object Detection (目标检测)

1. **R-CNN Family (Two-stage Detectors)**
   - **R-CNN**:
     - 1. Generate Region Proposals (e.g., Selective Search).
     - 2. Run CNN on **each** region (Very Slow!).
     - 3. Classify (SVM) & Regress Bbox.
   - **Fast R-CNN**:
     - **Improvement**: Run CNN on the **whole image** once.
     - Uses **RoI Pooling** to extract features for each proposal from the feature map.
     - Much faster than R-CNN.
   - **Faster R-CNN** (State-of-the-art):
     - **Improvement**: Replaces slow Selective Search with **RPN (Region Proposal Network)**.
     - RPN shares features with the detection network.
     - **Key Concept**: **Anchors** (reference boxes of various scales/ratios).
2. **YOLO (One-stage Detector)**
   - **YOLO (You Only Look Once)**:

- Treats detection as a **single regression problem**.
- Divides image into $S \times S$ grid. Each cell predicts Bbox coordinates (x, y, w, h) and class probabilities directly.
- **Pros**: Extremely fast (Real-time).
- **Cons**: Struggles with small objects or crowded scenes (compared to Faster R-CNN).

Coach's Past Paper 考法解析

必考 (Exam Focus - Past Paper Q6/Q7):

- **Faster R-CNN vs R-CNN**: Faster R-CNN uses **RPN** (learned proposals) instead of external algorithms, making it much faster.
- **YOLO vs Faster R-CNN**: YOLO is faster (one-stage) but less accurate for small objects. Faster R-CNN is more accurate (two-stage) but slower.
- **Scenario Question**: "Count food items" (Instance Segmentation) → Use **Mask R-CNN**. "Classify image pixels" → FCN.
- **FCN Upsampling**: Uses deconvolution/transposed convolution to get high-res output from low-res feature maps.

Part F: Feature Detection (特征检测) - CORE EXAM TOPIC

Harris Corner Detector (Harris 角点检测)

1. **Basic Idea (核心思想)**: A corner is a point where intensity changes significantly in **all directions** (向任何方向移动窗口，像素值都剧烈变化). *Intuition*:
   - **Flat (平坦)**: No change in any direction.
   - **Edge (边缘)**: Change in one direction only.
   - **Corner (角点)**: Significant change in all directions.
2. **Harris Matrix M (关键矩阵)**: Describes gradient distribution in a window.

$$M = \sum_{(x,y)\in W} w(x,y) \begin{bmatrix} I_x^2 & I_xI_y \\ I_xI_y & I_y^2 \end{bmatrix}$$

3. **Eigenvalue Analysis (特征值分析 - 必背判据)**: Let λ_1, λ_2 be eigenvalues of M.
   - **Corner**: Both λ_1 and λ_2 are large. (两个都大 → 角点)
   - **Edge**: One is large, one is small ($\lambda_1 \gg \lambda_2$ or vice versa). (一个大一个小 → 边缘)
   - **Flat**: Both are small. (两个都小 → 平坦)
4. **Response Function R (响应函数 - 避免算特征值)**:

$$R = \det(M) - k(\mathrm{trace}(M))^2 = \lambda_1\lambda_2 - k(\lambda_1 + \lambda_2)^2$$

   (k is constant, usually $0.04 - 0.06$)
   - $R >$ threshold (Large Positive): **Corner**.
   - $R < 0$ (Large Negative): **Edge**.
   - $|R| \approx 0$ (Small): **Flat**.
5. **Invariance Properties (选择题陷阱)**:
   - **Rotation Invariant**: **YES** (旋转后特征值大小不变).
   - **Translation Invariant**: **YES**.
   - **Scale Invariant**: **NO!** (放大后，尖锐的角点可能变成平滑的边缘，Harris 失效。这也是为什么需要 SIFT).

Coach's Past Paper 考法解析

计算题救命模版 (Harris Response Calculation):

题目：Given Harris matrix $M = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}$ (diagonal matrix), $k = 0.05$. Is it a corner?

解法: 1. **Identify** λ: For diagonal matrix, $\lambda_1 = 4, \lambda_2 = 1$. (非对角矩阵需算 Det 和 Trace)
2. **Calculate Det**: $\det(M) = 4 \times 1 = 4$.
3. **Calculate Trace**: $\mathrm{trace}(M) = 4 + 1 = 5$.
4. **Calculate R**: $R = \det - k(\mathrm{trace})^2 = 4 - 0.05 \times (5^2) = 4 - 1.25 = 2.75$.
5. **Conclusion**: $R = 2.75$ (Positive & Large) → **It is a Corner**.

SIFT (Scale Invariant Feature Transform)

Goal: Extract features invariant to **Scale**, **Rotation**, and Illumination (解决 Harris 不抗缩放的问题).

1. **Step 1: Scale-space Extrema Detection (找斑点)**:
   - **Method**: Use Gaussian Pyramid to find blobs across scales.
   - **Peak Finding**: Compare pixel with **26 neighbors** (8 current, 9 above, 9 below).
   - **Exam Point (DoG vs LoG)**: **LoG (Laplacian of Gaussian)** is theoretically best but slow. **DoG (Difference of Gaussian)** is used as a **fast approximation** of LoG.

$$DoG \approx (k-1)\sigma^2\nabla^2 G$$

2. **Step 2: Keypoint Localization (精定位)**: Refine location to sub-pixel accuracy. Discard **low contrast points** (noise) and **edge points**.
3. **Step 3: Orientation Assignment (定方向)**:
   - **Method**: Build a **36-bin** histogram of gradient directions ($360°/10° = 36$).
   - **Purpose**: Assign a dominant direction to achieve **Rotation Invariance** (抗旋转).
4. **Step 4: Keypoint Descriptor (生成身份证)**:
   - **Method**: Take 16×16 window → divide into 4×4 sub-regions.
   - Compute **8-bin** orientation histogram for each sub-region.
   - **Dimension (必考数字)**:

$$4 \times 4 \text{ (cells)} \times 8 \text{ (bins)} = \textbf{128}\text{-D vector}$$

   - **Normalization**: Normalize vector for **Illumination Invariance** (抗光照).

Feature Matching (特征匹配 - 老师强调)

1. **Naive Approach**: Euclidean Distance $\|f_1 - f_2\|$. **Problem**: Cannot distinguish ambiguous matches (e.g., fence posts).
2. **Lowe's Ratio Test (必考策略)**:
   - **Method**: Find the **Best Match** (nearest) and **2nd Best Match**.
   - **Formula**:

$$\text{Ratio} = \frac{\text{Distance(Best)}}{\text{Distance(2nd Best)}} < \text{Threshold (e.g., 0.8)}$$

   - **Logic**: If Ratio ≈ 1, it means the best match is not unique (ambiguous) → **Discard**. Only keep distinctive matches (Ratio small).

Coach's Past Paper 考法解析

Matching Example (题目做法):
Q: Best match distance = 10, 2nd best = 11. Threshold = 0.8. Keep it?
A: Ratio $= 10/11 \approx 0.9$. Since $0.9 > 0.8$, the match is ambiguous. **Reject/Discard**.

Other Concepts

1. **Blob Detection**: Uses **LoG** or **DoG** to find blob-like structures. (DoG is faster).
2. **Bag of Words (BoW)**:
   - Represents an image as a histogram of visual words (ignores position).
   - **Process**: SIFT → K-means (Codebook) → Quantize → Histogram.
   - **Spatial Pyramid Matching**: Adds spatial info back by dividing image into grids.

Part G: Transformations (图像变换) - CALCULATION HEAVY

Homogeneous Coordinates (齐次坐标 - 必考基础)

- **Why? (为什么要用)**: 普通的 2D 坐标 (x, y) 没法用矩阵乘法来表示 **Translation (平移)**。为了统一格式，强行加一个维度。
- **Conversion (转换规则)**:
  - **2D Point → Homogeneous**: $(x, y) \to [x, y, 1]^T$ (直接在屁股后面加个 1).
  - **Homogeneous → 2D Point**: $[x', y', w']^T \to (x'/w', y'/w')$.

Coach's Past Paper 考法解析

THE ULTIMATE TRAP (归一化陷阱 - 老师强调):
矩阵乘法算出来的结果通常是 $[u, v, w]^T$ (比如 $[300, 600, 3]^T$)。
千万别直接答 $(300, 600)$！**一定要除以最后一个数 w！**
正确答案是 $(300/3, 600/3) = (100, 200)$。如果不除，直接 0 分。

2D Transformations Hierarchy (2D 变换层级)

1. **Translation (平移)**: **2 DOF** (自由度: t_x, t_y). 形状方向都不变，只挪位置。
2. **Rigid / Euclidean (刚体变换)**: **3 DOF** (1 Rotation + 2 Translation). **Invariant**: Lengths (长度不变), Angles (角度不变). 就像把物体拿起来转个圈放下去，东西没变。
3. **Similarity (相似变换)**: **4 DOF** (Scale + Rigid). **Invariant**: Angles (角度不变). 大小变了，但形状没变。

4. **Affine (仿射变换 - 必考重点)**: **6 DOF**. (矩阵里有 6 个未知数 a, b, c, d, e, f).

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Invariant: **Parallelism (平行线性)**. (原来的平行线，变完还是平行的)。
Calculation: 需要 **3 pairs of points** (3 对点) 来求解。

5. **Projective / Homography (透视/单应性 - 最难)**: **8 DOF**. (矩阵里有 9 个数，但最后归一化少 1 个自由度，所以是 8). **Feature**: Parallel lines converge (平行线会相交，比如铁轨). **Calculation**: 需要 **4 pairs of points** (4 对点) 来求解。

Coach's Past Paper 考法解析

计算题模版: Solve for Affine Matrix (求仿射矩阵)
题目: Given 3 matches: $(x_1, y_1) \to (x'_1, y'_1)$, etc. Find matrix parameters.
解法: 不要去解矩阵乘法，直接拆成两个方程组解！

- 方程组 1 (求第一行 a, b, c): $x' = ax + by + c$ (把 3 个点的数值代进去解).
- 方程组 2 (求第二行 d, e, f): $y' = dx + ey + f$ (同理).

Past Paper 2019 Q3(b): "Projective transform of a line" → answer is $H^{-T}l$ (直接背结论: 点变换用 H, 线变换用 H 的逆转置).

Image Warping (图像扭曲 - 怎么搬运像素)

1. **Forward Warping (前向映射 - 坏方法)**:
   - **做法**: 把源图像素 (x, y) 算到新位置 (x', y')，把颜色填进去。
   - **Problem**: **Holes/Gaps (会有洞)**。因为放大图片时，像素就像被拉开的网，中间有空隙。
2. **Inverse Warping (逆向映射 - 老师钦定好方法)**:
   - **做法**: 遍历目标图的每个像素 (x', y')，反向算出它在原图的坐标 $(u, v) = H^{-1}(x', y')$。
   - **Problem**: 算出来的 (u, v) 通常是小数 (例如 3.6, 4.2)，原图没有坐标是小数的像素。
   - **Solution (必考填空)**: Must use **Interpolation (插值)**! (比如 Bilinear 双线性插值，用周围 4 个格子算).

Image Alignment (RANSAC - 怎么剔除坏点)

当匹配点里有很多 **Outliers (误匹配/坏点)** 时：

- **Least Squares (最小二乘法)**: **Bad**. 它试图照顾所有点，一个离谱的坏点就能把整个模型拉偏 (Sensitive to outliers)。
- **RANSAC (随机采样一致性 - Good)**: **Robust**. 专门用来对付 Outliers。**Steps (背诵流程)**: 1. Randomly sample minimal points (e.g. Affine 取 3 个，Homography 取 4 个). 2. Fit model (算出矩阵). 3. Count **inliers** (数数有多少个点支持这个模型). 4. Repeat & pick the model with most inliers.

Coach's Past Paper 考法解析

必考计算题: RANSAC Iterations (迭代次数)
题目: Outlier ratio $\epsilon = 40\%$ (0.4), sample size $s = 3$. Want 99% confidence ($p = 0.99$). How many iterations N?
公式 (必背):

$$N = \frac{\log(1-p)}{\log(1-(1-\epsilon)^s)}$$

步骤: 1. 算分母内部: $(1-\epsilon)^s = (1-0.4)^3 = 0.6^3 = 0.216$ (这是全抽到好点的概率). 2. 算分母: $\log(1-0.216) = \log(0.784) \approx -0.105$. 3. 算分子: $\log(1-0.99) = \log(0.01) \approx -2$. 4. 除法: $N = -2/-0.105 \approx 19.04$. 5. **结论**: Round up to **20** iterations. (必须向上取整!).

Part H: Camera Models (相机模型) - CALCULATION CORE

Pinhole Camera Model (针孔相机模型)

1. **Coordinate Systems (坐标系)**
   - **World Coordinate** (X_w, Y_w, Z_w): Arbitrary origin.
   - **Camera Coordinate** (X_c, Y_c, Z_c): Origin at camera center (pinhole), Z-axis along optical axis.
   - **Image Coordinate** (x, y): Physical 2D plane (e.g., film sensor).
   - **Pixel Coordinate** (u, v): Digital image indices.
2. **Projection Steps (投影步骤)**
   (a) **World to Camera (Extrinsics)**:

   $$P_c = [R|T]P_w$$

   R is 3×3 rotation, T is 3×1 translation.

   (b) **Camera to Image (Perspective)**: Using similar triangles: $x = f\frac{X_c}{Z_c}$, $y = f\frac{Y_c}{Z_c}$. In homogeneous form:

   $$\lambda \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X_c \\ Y_c \\ Z_c \\ 1 \end{bmatrix}$$

   (c) **Image to Pixel (Intrinsics)**: Convert physical units (mm) to pixels.

   $$u = \alpha x + c_x, \quad v = \beta y + c_y$$

   where α, β are scaling factors (pixels/mm), (c_x, c_y) is the principal point (center).
3. **Full Projection Matrix (完整投影矩阵)**

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K[R|T] \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} = MP_w$$

where K is the **Intrinsic Matrix**:

$$K = \begin{bmatrix} \alpha & s & c_x \\ 0 & \beta & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

(s is skew, usually 0).

Exam Calculation Guide (计算指南)

1. **Calculation Steps (必背)**
   - **Step 1**: Write down $P_w = [X, Y, Z, 1]^T$.
   - **Step 2**: Multiply by Extrinsic Matrix $[R|T]$ to get Camera Coords.
   - **Step 3**: Multiply by Intrinsic Matrix K.
   - **Step 4**: Result is a vector $P' = [u', v', w']^T$.

Coach's Past Paper 考法解析

必考陷阱 (The Normalization Trap):

- The result from matrix multiplication is **Homogeneous**: $[u', v', w']^T$.
- **CRITICAL**: The pixel coordinate is NOT (u', v').
- You MUST divide by the last element w' (depth/scaling factor):

$$u = u'/w', \quad v = v'/w'$$

- Example: If result is $[300, 450, 3]^T$, the pixel is $(100, 150)$. Teacher: "If you forget this, you lose points!"

2. **Parameters Counting**
   - **Extrinsics**: 6 DOF (3 Rotation, 3 Translation).
   - **Intrinsics**: 5 DOF (f_x, f_y, c_x, c_y, s).
   - **Projection Matrix M**: 3×4 matrix, 11 DOF (defined up to scale).

Part H Supplement: 核心解题指南 (Survival Guide)

Coordinate Transformation Pipeline (坐标变换流水线)

想象一个点从现实世界跑到你的照片上，要过三关：

(a) **World → Camera (世界到相机)**:
- 靠 **Extrinsic Matrix (外参)** $[R|t]$。
- 作用：把世界坐标系旋转平移，对齐到相机坐标系。
- **DOF**: 6 (3 个转动角度 + 3 个位移距离).

(b) **Camera → Image Plane (相机到物理成像面)**:
- 靠 **Perspective Projection (透视投影)**。
- 公式：$x = f\frac{X}{Z}, y = f\frac{Y}{Z}$。
- 原理：**Similar Triangles (相似三角形)**。离得越远 (Z 越大)，像越小。

(c) **Image Plane → Pixel (物理面到像素)**:
- 靠 **Intrinsic Matrix (内参)** K。
- 作用：把毫米 (mm) 单位转换成像素 (pixel) 单位，并加上原点偏移。
- **DOF**: 5 (f_x, f_y 焦距, c_x, c_y 中心点, s 倾斜).

Coach's Past Paper 考法解析

计算题必背模版 (Projection Calculation):
题目: Given World Point P_w, Extrinsics $[R|t]$, Intrinsics K. Find pixel (u, v).
Step 1: Matrix Multiplication (矩阵相乘)

$$\begin{bmatrix} u' \\ v' \\ w' \end{bmatrix} = K \cdot [R|t] \cdot \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix}$$

(注意：算出来的是一个 3 维向量，最后一位 w' 通常不为 1).
Step 2: Normalization (归一化 - 必做!)
Divide by the last element w'!

$$u = u'/w', \quad v = v'/w'$$

Example: Result is $[3000, 4500, 10]^T$.
Pixel coordinate = $(3000/10, 4500/10) = (\mathbf{300}, \mathbf{450})$.
(If you forget to divide, you get 0 points!)

Important Concepts (必考概念)

- **Pinhole Model**: Ideal camera. All rays pass through a single point (Center of Projection). No lens distortion assumed.
- **Principal Point** (c_x, c_y): The intersection of the **Optical Axis** and the Image Plane. Usually the image center.
- **Focal Length** f: Distance between Pinhole and Image Plane.

Part H: Camera Models (相机模型) - 备考与使用指南

1. 核心概念讲解 (Coach's Breakdown)

这部分内容是整门课中 **计算量最大、逻辑最死板**的部分。只要你掌握了“坐标变换流水线”和“齐次坐标归一化”，这里就是送分题。

核心逻辑：从世界到像素的旅程

想象一个点 P 从现实世界跑进你的照片里，它必须经过三道关卡：

1. **第一关：World → Camera (世界坐标系 → 相机坐标系)**
   - **工具**：外参矩阵 (Extrinsic Matrix) $[R|t]$。
   - **作用**：把世界坐标系旋转 (R) 并平移 (t)，让相机的光心成为原点，光轴指向 Z 轴正方向。
   - **考点**：外参有 6 个自由度 (3 个旋转角 + 3 个位移分量)。
2. **第二关：Camera → Image Plane (相机坐标系 → 物理成像平面)**
   - **工具**：透视投影 (Perspective Projection)。
   - **公式**：$x = f\frac{X}{Z}, y = f\frac{Y}{Z}$。
   - **直觉**：**近大远小**。离相机越远 (Z 越大)，成像越小。这是由相似三角形原理决定的。
3. **第三关：Image Plane → Pixel (物理成像平面 → 像素坐标系)**
   - **工具**：内参矩阵 (Intrinsic Matrix) K。
   - **作用**：
     (a) 单位转换：把物理单位 (mm) 转换成像素单位 (pixel)。需要乘以缩放因子 α, β。
     (b) 原点平移：物理成像平面的原点在中心，像素坐标系的原点在左上角。需要加上偏移量 c_x, c_y。
   - **考点**：内参有 5 个自由度 (f_x, f_y, c_x, c_y, s)。

齐次坐标 (Homogeneous Coordinates)

为了把加法（平移）写进矩阵乘法里，我们需要给坐标加一个维度（通常加 1)。

- 欧氏坐标 $(x, y) \to$ 齐次坐标 $[x, y, 1]^T$。
- **关键点**：齐次坐标具有缩放不变性，$[x, y, 1]^T$ 和 $[kx, ky, k]^T$ 代表同一个点。这就是为什么最后一步要做“归一化”。

2. 必考题型与 Cheat Sheet 使用方法

题型一：投影计算大题 (The Big Calculation)

考法：题目给你一个世界坐标点 $P_w = (X, Y, Z)$，给你内参 K 和外参 R, T，让你求图片上的像素坐标 (u, v)。这是送分大题，通常占 10-15 分。
如何使用 Cheat Sheet：

1. 直接定位到 **Section 2: Exam Calculation Guide**。
2. **Step 1**：把题目给的世界坐标写成 4 维齐次形式：$P_w = [X, Y, Z, 1]^T$。
3. **Step 2**：列出矩阵连乘公式：$P' = K[R|t]P_w$。
4. **Step 3 (陷阱)**：算出结果是一个 3 维向量 $[u', v', w']^T$。此时千万不要直接写答案！
5. **Step 4 (必做)**：**归一化 (Normalization)**。把前两个数除以第三个数 w'。

$$u = u'/w', \quad v = v'/w'$$

6. **检查**：如果 w' 是深度 Z，它通常是一个比较大的数。如果不除以它，算出来的像素坐标会非常大（例如几千几万），这显然不对。

题型二：参数自由度 (DOF Counting)

考法：选择题或填空题，问你外参、内参或投影矩阵有多少个自由度。
如何使用 Cheat Sheet：

- 直接看 **Section 2 Item 2: Parameters Counting**。
- **Extrinsics**: 6 (3 Rotation + 3 Translation).
- **Intrinsics**: 5 (f_x, f_y, c_x, c_y, s).
- **Projection Matrix M**: 11 (虽然是 3×4 矩阵有 12 个元素，但因为齐次坐标的缩放不变性，少一个自由度)。

题型三：逆向推导 (Reverse Engineering)

考法：(参考 2020 Q11) 给你投影矩阵 M，问相机在哪里？或者给你图像点，让你求世界点（通常需要多视角）。
解题思路：

- 相机中心 C 在世界坐标系中满足 $MC = 0$。
- 简单的平移关系：$t = -RC$ (即 $C = -R^T t$)。

3. 考前最后叮嘱

- **死记硬背矩阵顺序**：$P_{pixel} = K \times [R|t] \times P_{world}$。顺序不能反！先内参，再外参，最后世界坐标。
- **单位换算**：如果题目给的焦距是毫米 (mm)，记得看看有没有给缩放因子 (pixels/mm)。如果有，要先乘起来得到像素单位的焦距 (f_x, f_y)。
- **归一化是拿分关键**：再次强调，算出矩阵乘积后，**一定要除以第三个元素**。这是老师强调的“区分优秀学生”的细节。

Part I: Stereo Vision (立体视觉)

Depth from Stereo (双目测距)

1. **Basic Setup (Standard Case)**
   - Two identical cameras with parallel optical axes.
   - Separated by baseline B (horizontal).
   - Focal length f.
2. **Disparity Formula (必考公式)**
   - Let x_l be x-coord in Left Image, x_r in Right Image.
   - **Disparity (视差)**: $d = x_l - x_r$.
   - **Depth Calculation**:

$$Z = \frac{f \cdot B}{d}$$

   - **Key Relation**: Depth Z is **inversely proportional** to Disparity d.
   - Near objects → Large disparity.
   - Far objects → Small disparity.

Epipolar Geometry (极线几何)

1. **Key Concepts**
   - **Epipole (极点)** e: Intersection of baseline with image plane. Projection of other camera center.
   - **Epipolar Line (极线)** l: Intersection of epipolar plane with image plane. All epipolar lines intersect at the epipole.
   - **Epipolar Constraint**: Given point p in Left Image, its match p' in Right Image MUST lie on the corresponding epipolar line l'. (Reduces search from 2D to 1D).
2. **Image Rectification (图像校正)**
   - **Goal**: Warp images so that epipolar lines became **horizontal and parallel** (scanlines).
   - **Result**: Epipoles move to infinity. Disparity only in x-direction ($y_l = y_r$).
   - Makes matching efficient (scanline search).

Coach's Past Paper 考法解析

必考 (Exam Focus - Past Paper Q4):

- **Rectification Why?**: To simplify matching (1D search along horizontal lines).
- **Epipole at Infinity**: After rectification, epipoles are at infinity $[1, 0, 0]^T$.
- **Fundamental Matrix**: For rectified images, F has a special form (skew-symmetric).

Stereo Correspondence (立体匹配)

1. **Matching Methods**
   - **Window-based**: Compare small windows around pixels.
   - **Metrics**:
     - SSD (Sum of Squared Differences).
     - SAD (Sum of Absolute Differences).
     - NCC (Normalized Cross Correlation) - Robust to lighting changes.
2. **Challenges**
   - **Textureless regions**: Aperture problem, ambiguous matches.
   - **Repetitive patterns**: Multiple good matches.
   - **Occlusion**: Points visible in only one camera.

Part J: Structure from Motion (运动恢复结构)

Concepts (基本概念)

1. **Problem Statement**
   - **Given**: M images of a static scene.
   - **Goal**: Recover:
     (a) **Structure**: 3D coordinates of scene points (X_j).
     (b) **Motion**: Camera pose for each image (R_i, T_i).
   - **Ambiguity**: Solution is up to a global scale factor (cannot determine absolute size).
2. **Point Cloud (点云)**
   - A collection of **un-ordered** points.
   - Attributes: Geometry (x, y, z), Color (r, g, b).
   - Difference from Mesh: No connectivity/topology info.

Bundle Adjustment (光束法平差)

1. **Objective Function (必考)**
   - Minimize the sum of squared **Reprojection Errors**:

$$E = \sum_{i=1}^{M}\sum_{j=1}^{N} w_{ij}||P(X_j, R_i, T_i) - x_{ij}||^2$$

   - $P(\ldots)$: Predicted pixel coordinate (using camera model).
   - x_{ij}: Observed pixel coordinate (from feature matching).
   - w_{ij}: Visibility indicator (1 if point j visible in image i, else 0).
2. **Optimization**
   - Problem is **Non-linear Least Squares**.
   - Algorithm: **Levenberg-Marquardt (LM)**.
   - Needs good initialization to avoid local minima.

Coach's Past Paper 考法解析

必考 (Exam Focus - Past Paper Q8):
- **Chicken-and-Egg Problem**: Need 3D points to get Camera Pose; Need Camera Pose to get 3D points. SfM solves both simultaneously.
- **Counting Unknowns**: M images, N points. Unknowns: $6M$ (Cameras) + $3N$ (Points). Constraints: $2MN$ (2 eqs per observed point). Condition: $2MN \geq 6M + 3N - 7$ (minus 7 for global gauge freedom).

Part K: Motion & Optical Flow (光流法) - MAJOR

Optical Flow Equation (光流方程 - 基础理论)

1. **Three Key Assumptions (必考三大假设)**:
   - **1. Brightness Constancy (亮度恒定)**:

$$I(x, y, t) = I(x+u, y+v, t+1)$$

     (Object moves, but intensity stays same. 像素值随物体移动保持不变。)
   - **2. Small Motion (微小运动)**: Points move very little, allowing Taylor Expansion. (泰勒展开的前提)
   - **3. Spatial Coherence (空间一致性)**: Neighbors move together. (Used to solve Aperture Problem).
2. **Derivation (方程推导 - 简答题)**:
   - Taylor Expansion: $I(x+u, y+v, t+1) \approx I(x, y, t) + I_x u + I_y v + I_t$
   - Apply Constancy $(I(t+1) - I(t) = 0)$:

$$\mathbf{I_x u + I_y v + I_t = 0} \quad \Rightarrow \quad \nabla I \cdot [u, v]^T = -I_t$$

   - **Meaning**: Spatial Gradient × Velocity + Temporal Change = 0.

Lucas-Kanade Algorithm (LK 算法 - 求解平移)

1. **Aperture Problem (孔径问题)**:
   - **Issue**: 1 pixel gives 1 eq $(I_x u + I_y v = -I_t)$ but 2 unknowns (u, v). **Solution is not unique**.
   - **Visual**: Seeing through a small hole, cannot determine motion along the edge. (沿边缘运动无法观测)
   - **Fix**: Assume a window (e.g., 5×5) has **constant motion** (u, v).
2. **Least Squares Solution (最小二乘解 - 计算核心)**: Stack equations for window size N (e.g., 25):

$$\underbrace{\begin{bmatrix} I_{x1} & I_{y1} \\ \vdots & \vdots \\ I_{xN} & I_{yN} \end{bmatrix}}_{A(N\times 2)} \begin{bmatrix} u \\ v \end{bmatrix} = -\underbrace{\begin{bmatrix} I_{t1} \\ \vdots \\ I_{tN} \end{bmatrix}}_{\mathbf{b}(N\times 1)} \quad \Rightarrow \quad A\mathbf{d} = \mathbf{b}$$

Solution: $\mathbf{d} = (A^TA)^{-1}A^T\mathbf{b}$. (Requires A^TA invertible).

3. **Eigenvalues & Reliability (特征值物理意义)**: Analyze eigenvalues λ_1, λ_2 of matrix $H = A^TA$:
   - **Corner**: Large λ_1, λ_2. (Good for tracking, invertible). BEST Feature.
   - **Flat**: Small λ_1, λ_2. (No texture, matrix empty/noise).
   - **Edge**: $\lambda_1 \gg \lambda_2$. (Aperture problem, singular matrix).

Advanced Topics (高分必考大题)

Affine Optical Flow (仿射光流 - Past Paper Q14)

Scenario: Motion is NOT pure translation, but Affine (rotation, scale, shear):

$$u(x, y) = ax + by + c, \quad v(x, y) = dx + ey + f$$

Goal: Solve for **6** unknowns (a, b, c, d, e, f).

Coach's Past Paper 考法解析

Step-by-Step Derivation (满分步骤):
1. **Plug into Optical Flow Eq**: $I_x u + I_y v = -I_t$
2. **Substitute** u, v: $I_x(ax + by + c) + I_y(dx + ey + f) = -I_t$
3. **Regroup variables**:

$$(I_x x)a + (I_x y)b + (I_x)c + (I_y x)d + (I_y y)e + (I_y)f = -I_t$$

4. **Matrix Form (1 pixel)**:

$$[I_x x,\ I_x y,\ I_x,\ I_y x,\ I_y y,\ I_y] \cdot [a, b, c, d, e, f]^T = -I_t$$

5. **Constraint**: Need at least **3 pixels** (6 eqs) to solve 6 unknowns. Usually use window.

Large Motion Problem (大位移问题)

- **Problem**: LK fails if motion > 1 pixel (violates Taylor expansion/Small Motion).
- **Solution**: **Image Pyramids (Coarse-to-fine)**.
- **Process**: 1. Downsample (Make image small → motion becomes small). 2. Solve LK at coarse level. 3. **Upsample** flow and **Warp** image. 4. Refine at fine level (Iterative).

Coach's Past Paper 考法解析

Teacher's Trap / Critical Detail: When performing **Image Warping** (shifting image by flow u, v), the new coordinates (x', y') are float numbers (non-integers). **You MUST use Interpolation** (e.g., Bilinear/Bicubic) to compute pixel values. *Answer "Interpolation" if asked "What is crucial for warping?".*

Part K: Motion & Optical Flow (光流法) - 备考与使用指南

1. 核心概念讲解 (Coach's Breakdown)

这部分内容在考试中属于 **Major Topic (重点)**，通常会涉及推导题（大题）和概念辨析题（简答题）。

核心逻辑：Brightness Constancy (亮度恒定)

这是光流法的灵魂。

- **直觉**：一个物体在运动，但它身上的“颜色/亮度”是不会变的。比如我的红衣服，走到哪里都是红的。
- **公式**：$I(x,y,t) = I(x+u, y+v, t+1)$
- **推导结果**：$I_x u + I_y v + I_t = 0$。这个方程告诉你：**空间梯度 (I_x, I_y) 和时间变化 (I_t) 是如何通过速度 (u, v) 联系起来的**。

LK 算法 (Lucas-Kanade) 与孔径问题 (Aperture Problem)

- **孔径问题**：想象你透过一根吸管（小孔）看世界。如果一根白墙上的黑线在移动，你只能看到线在“动”，但你不知道它是顺着线滑，还是垂直移动。
- **数学本质**：一个像素只能提供一个方程，但有两个未知数 (u, v)，解不出来。
- **LK 的解法**：“众筹”。假设我周围 5×5 的邻居都跟我做一样的运动。这样我就有 25 个方程来解 2 个未知数，这就变成了 **最小二乘问题 (Least Squares)**。

矩阵 $A^T A$ 与特征值 (Eigenvalues)

我们在 Part F (Harris Corner) 里学过这个矩阵。

- **Corner (角点)**：两个特征值都大。就像拼图的直角边，往哪儿移都能看出来，所以**最适合追踪**。
- **Flat (平坦区域)**：特征值都接近 0。就像一张白纸，怎么移都看不出来，**追踪失败**。
- **Edge (边缘)**：一个大一个小。这就是孔径问题，只能确定一个方向的运动。

大位移问题 (Large Motion)

- **为什么 LK 会失败？**因为泰勒展开只在 Δx 很小时才准。如果物体“瞬移”了很远，泰勒展开就失效了。
- **解法**：**金字塔 (Pyramids)**。先把图缩得很小（下采样），那么原本的大位移在小图上就变成了小位移。在小图上算好大概方向，再回到大图上精修。

2. 必考题型与 Cheat Sheet 使用方法

题型一：推导题 (Derivation)

考法：题目会给出一个非标准的运动模型，比如 **Affine Motion (仿射运动)**，让你写出求解的线性方程组。

如何使用 Cheat Sheet：

1. 直接定位到 **3.1 Affine Optical Flow** 的 `coachbox`。
2. **照抄步骤**：
   - 写出标准光流方程：$I_x u + I_y v = -I_t$
   - 把题目给的 u, v 公式（例如 $u = ax + by + c$）代进去。
   - 把未知数（$a, b, c...$）整理到一边，已知量（I_x, I_y, I_t）整理到另一边。
   - 写成矩阵形式 $A\mathbf{x} = \mathbf{b}$。
3. **得分点**：最后一定要写出矩阵的具体形式，并说明需要多少个像素（方程数 $\geq$ 未知数）才能解。

题型二：概念简答 (Short Questions)

考法：

- ”What are the assumptions of optical flow?”
- ”Why do we need image pyramids?”
- ”Why is corner better than edge for tracking?”

如何使用 Cheat Sheet：

- **假设**：直接抄 Section 1 的”Three Key Assumptions”。
- **金字塔**：直接抄 Section 3.2 的”Large Motion Problem”。
- **特征好坏**：直接抄 Section 2.3 的”Eigenvalues & Reliability”。

题型三：陷阱题 (Details)

考法：”When implementing iterative LK with pyramids, what implies regarding image resampling?” (在使用金字塔 LK 时，图像重采样有什么要求？)

如何使用 Cheat Sheet：

- 看最后的红色 `coachbox`。
- **标准答案**：**Interpolation (插值)** is required. 因为 Warp 之后的位置 (x', y') 通常是小数，数字图像没有小数坐标，必须用双线性插值 (Bilinear Interpolation) 算出像素值。

3. 考前最后叮嘱

- **分清 u, v 和 x, y**：
  - x, y 是像素的位置坐标（已知的）。
  - u, v 是我们要解的速度（未知的）。
  - 在推导矩阵时，要把 u, v 提出来作为未知数向量。
- **RANSAC 的联系**：
  - 如果在计算光流或者图像变换时有 **Outliers (异常值/错误匹配)**，记得提 **RANSAC**。这是万能的“鲁棒性”答案。
- **计算器**：
  - 虽然是推导为主，但如果考到 2×2 矩阵求逆或者特征值，记得带好计算器。